

Vehicle Loan Default Prediction

A Credit Risk Model Using Real LTFS (L&T Financial Services) NBFC Data

1. Objective

This project builds a predictive model to estimate the probability that a borrower will default on a vehicle loan, using real historical loan data from LTFS (L&T Financial Services), one of India's leading NBFCs. The goal is to identify which borrower and loan characteristics are most predictive of default, and to evaluate how well a data-driven model can support underwriting decisions.

2. Data

The dataset contains 233,154 vehicle loans with 41 fields covering loan terms, borrower demographics, document verification, and credit bureau history. The overall default rate in the dataset is 21.7%, meaning roughly 1 in 5 loans in the sample defaulted.

A key data quality finding: 116,950 loans (50.2% of the dataset) show “No Bureau History Available” — a common and important real-world pattern in the Indian lending market, where a large share of borrowers are new to formal credit. This was treated as meaningful information (via a dedicated “has bureau score” flag) rather than dropped or treated as a numerically low score.

3. Feature Engineering

Several raw fields required transformation before modeling:

- Date of Birth and Disbursal Date were converted into borrower age at disbursal.
- Text fields such as “1yrs 11mon” (average account age, credit history length) were parsed into total months.
- Bureau score was split into two features: a binary “has bureau score” flag, and the score itself (populated only where a real score exists), to avoid the model misinterpreting “no history” as “worst score.”
- Employment type was converted into a binary self-employed indicator, with missing values kept as a separate “Unknown” category rather than dropped.

18 features spanning loan terms, credit bureau history, and borrower profile were used in the final models.

4. Methodology

Two models were trained and compared:

- Logistic Regression — the industry-standard baseline for credit risk scoring, valued for its interpretability.
- Random Forest — a non-linear ensemble model, used to check whether more complex feature interactions meaningfully improve on the linear baseline.

The data was split 80/20 into training and test sets, stratified to preserve the 21.7% default rate in both sets. Given the class imbalance (only ~22% of loans default), models were evaluated using AUC-ROC and precision/recall rather than raw accuracy — a model predicting “no default” for every loan would achieve 78% accuracy while catching zero actual defaults, making accuracy a misleading metric for this problem.

5. Results

Model	AUC-ROC
Logistic Regression (baseline)	0.613
Random Forest	0.626

Random Forest improved on the logistic regression baseline by 1.3 percentage points of AUC — a modest but real gain, indicating the relationships in this data are largely linear, with a small amount of additional non-linear signal captured by the ensemble model. An AUC in the 0.61–0.63 range is consistent with published benchmarks on this dataset; even heavily tuned

models in public competitions on this same data typically reach only 0.66–0.68, reflecting genuine noise in real-world credit data rather than a modeling shortfall.

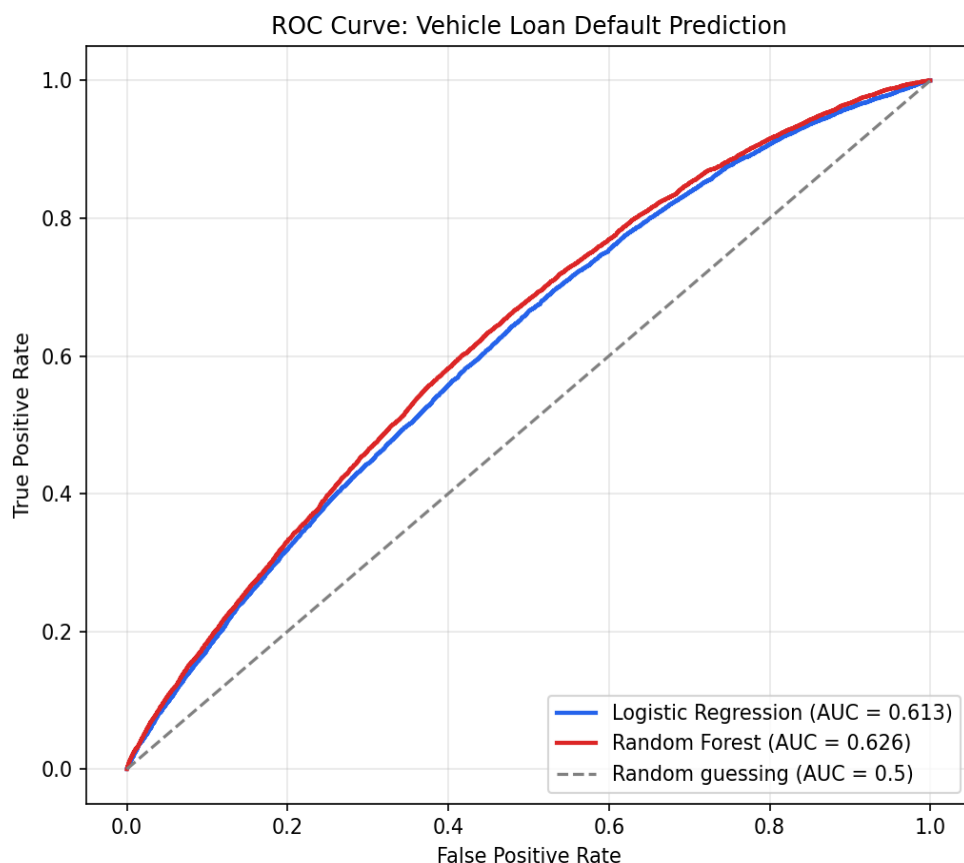


Figure 1: ROC curves for both models. Random Forest (red) sits consistently above Logistic Regression (blue), confirming a small but consistent improvement.

5.1 Precision / Recall Trade-off

At the default 0.5 classification threshold, the logistic regression model achieved 27% precision and 63% recall on the “default” class. In practical terms: of loans flagged as high-risk, only about 1 in 4 actually defaulted, but the model successfully caught roughly 2 out of every 3 loans that did default. This reflects a genuine business trade-off familiar to credit risk teams — a stricter threshold reduces false positives (rejecting good customers) at the cost of missing more actual defaults, and vice versa. The appropriate threshold depends on the lender's relative cost of a missed default versus a rejected good customer.

6. Key Drivers of Default Risk

Both models independently agreed on the three strongest predictors of default:

- Loan-to-Value (LTV) ratio — the single strongest predictor in both models. Higher LTV (less borrower equity in the financed asset) is consistently associated with higher default risk.
- Disbursed loan amount — larger loans carry higher default risk.
- Bureau score — where available, a higher bureau score is protective against default, as expected.

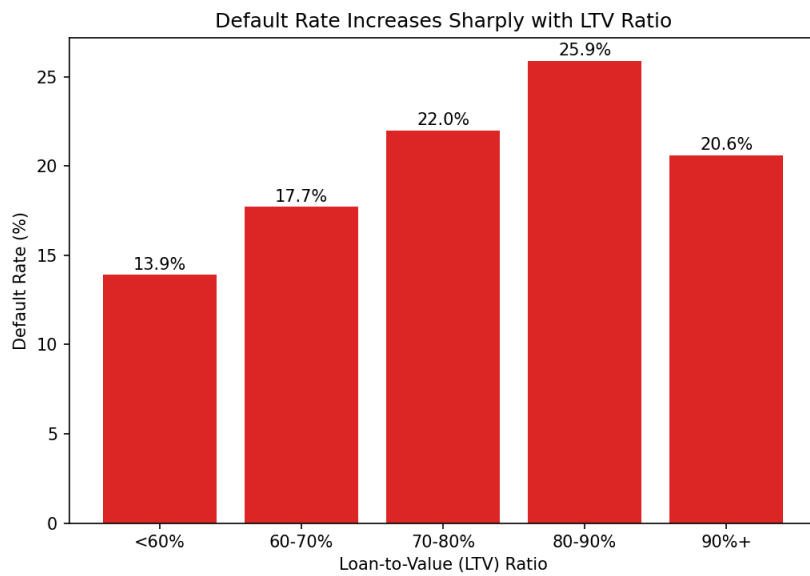


Figure 2: Default rate by LTV bucket. Default risk rises from 13.9% (LTV under 60%) to 25.9% (LTV 80–90%), a clear and intuitive monotonic relationship.

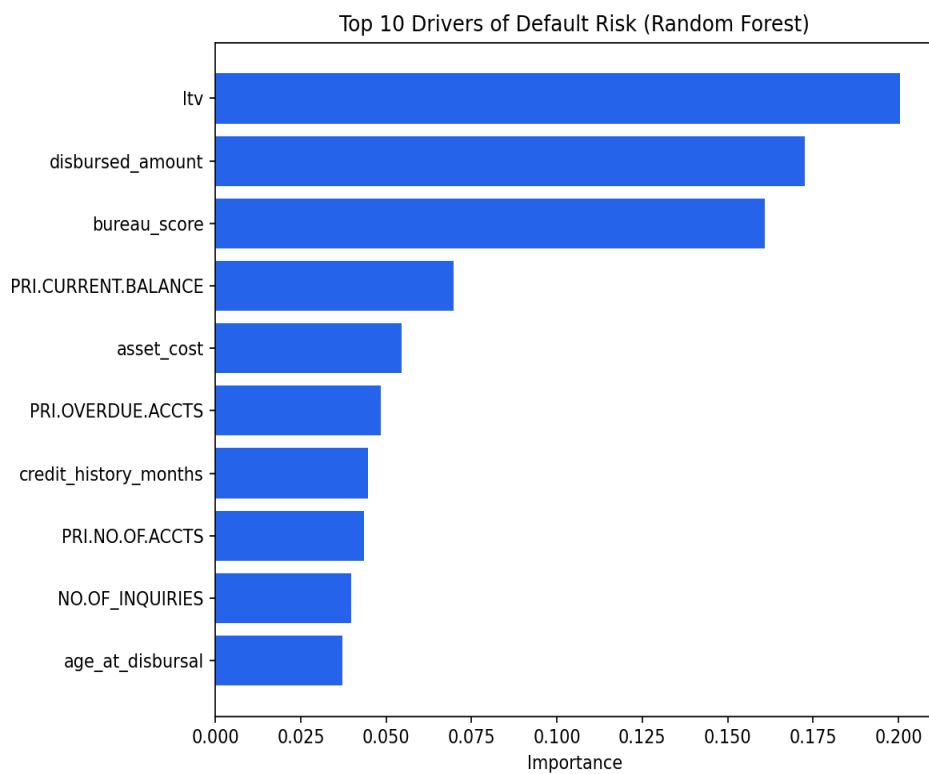


Figure 3: Top 10 default risk drivers by Random Forest feature importance.

7. Business Interpretation

The LTV finding has a direct, actionable interpretation for underwriting: borrowers financing a larger share of the asset's value (i.e., putting down less of their own money) are meaningfully more likely to default. This supports tightening LTV caps or applying risk-based pricing at higher LTV bands, a lever most NBFCs already use but one this analysis quantifies with real data.

The prevalence of “no bureau history” borrowers (over 50% of the portfolio) also has a strategic implication: a large share of the loan book cannot be effectively risk-differentiated using traditional bureau data alone. This is precisely the segment where alternative data (utility payments, mobile usage patterns, etc.) is increasingly used by Indian fintechs and NBFCs to improve underwriting.

8. Limitations

- Class imbalance (21.7% default rate) means model performance metrics must be read as AUC/precision/recall rather than accuracy; accuracy alone would be misleading.
- Missing bureau history for 50% of borrowers limits the model's ability to differentiate risk within that segment; a production model would likely incorporate alternative data sources.
- Both models are baseline implementations (no hyperparameter tuning or advanced feature engineering such as interaction terms); AUC could likely be improved further with additional tuning, as seen in public leaderboard results on this dataset.
- This is a first-EMI default definition (whether the borrower defaults on the very first payment) rather than default over the full loan tenure, which is a narrower and somewhat different risk question than lifetime default probability.

9. Conclusion

This project builds an interpretable, real-data credit risk model that identifies loan-to-value ratio, loan size, and bureau credit history as the primary drivers of vehicle loan default risk for an Indian NBFC. While predictive performance (AUC ~0.61 – 0.63) reflects the genuine difficulty of this dataset rather than a modeling shortcoming, the findings are directionally clear, consistent across two different model types, and actionable for underwriting policy — particularly around LTV-based risk pricing and the treatment of credit-invisible borrowers.

10. Data Source

- LTFS Vehicle Loan Default Prediction dataset, originally released for the L&T Financial Services “Data Science FinHack” competition (via Kaggle).